

Cache Management

Table of Contents

- Cache Management
 - Purpose
 - Cache Strategy
 - Application Memory Cache
 - Invalidation Rules
 - HTTP Header Evidence
 - Automated Evidence

Cache Management

Purpose

The system uses controlled cache management to improve repeated read performance without exposing private academic data. Cache behavior is explicit at three layers: HTTP response headers, an application memory cache with TTL and tag invalidation, and Nginx static asset rules for the deployed frontend.

Cache Strategy

Area	Policy	Evidence
Sensitive API responses	Cache-Control: no-store, no-cache, must-revalidate, private	X-Cache-Policy: sensitive-no-store
Public auth configuration	Cache-Control: public, max-age=3600, must-revalidate	GET /api/v1/auth/config

Area	Policy	Evidence
Reference catalogs	Private HTTP cache plus memory cache	GET /api/v1/roles , GET /api/v1/permissions
Stable academic lists	Private HTTP cache plus memory cache	GET /api/v1/branches , GET /api/v1/dance-categories , GET /api/v1/dance-styles
Branch summary report	Private scoped memory cache, 30 second TTL	GET /api/v1/reports/branches/summary
Static frontend assets	Long-lived immutable cache	Nginx /assets/ and Express static headers
HTML and login pages	Revalidate on every navigation	Cache-Control: no-cache, must-revalidate
Python analytics health	Public short cache, 60 second TTL	GET /api/analytics/v1/health
Protected Python analytics	No-store	X-Cache-Policy: sensitive-no-store

Application Memory Cache

Node.js creates a shared `CacheService` when the Express app starts. The cache stores in-process entries with:

- Key-based lookup.
- TTL expiration.
- Tag-based invalidation.
- Actor-scoped keys for role and branch-sensitive data.
- Response evidence headers: `X-Memory-Cache` , `X-Memory-Cache-Key` and `X-Memory-Cache-TTL` .

The current memory-cached resources are:

Resource	TTL	Cache Scope	Invalidated By
Roles catalog	1800 seconds	Shared internal catalog	Application restart
Permissions catalog	1800 seconds	Shared internal catalog	Application restart

Resource	TTL	Cache Scope	Invalidated By
Branch list	300 seconds	Authenticated actor role/id	Branch changes and branch access changes
Dance categories	600 seconds	Authenticated actor role/id	Category changes
Dance styles	600 seconds	Authenticated actor role/id	Style changes
Branch summary report	30 seconds	Authenticated actor role/id	Branch, student, attendance, justification, scholarship and promotion changes

Actor-scoped cache keys prevent a BranchDirector response from being reused for another user with different branch access.

Invalidation Rules

State-changing academic actions invalidate related tags immediately. Examples:

- Creating or updating a branch invalidates `branches` and `reports`.
- Creating or updating a student invalidates `students` and `reports`.
- Recording student attendance invalidates `attendance` and `reports`.
- Creating or reviewing an absence justification invalidates `attendance`, `justifications` and `reports`.
- Creating scholarship evaluations invalidates `evaluations` and `reports`.
- Creating level promotion evaluations invalidates `evaluations`, `students` and `reports`.
- Updating branch access invalidates `branches`, `students`, `teachers` and `reports`.

This keeps cache entries short-lived and removes affected entries as soon as business data changes.

HTTP Header Evidence

Manual verification can be done with:

```
curl -k -I https://18-217-255-109.sslip.io/api/v1/auth/config
curl -k -I https://18-217-255-109.sslip.io/api/v1/auth/me
curl -k -I https://18-217-255-109.sslip.io/
curl -k -I https://18-217-255-109.sslip.io/assets/<built-asset-file>
```

Expected evidence:

- Public config includes `Cache-Control: public, max-age=3600, must-revalidate`.
- Sensitive session routes include `Cache-Control: no-store`.
- HTML includes `X-Cache-Policy: html-revalidate`.
- Versioned frontend assets include `Cache-Control: public, max-age=31536000, immutable`.

Authenticated memory cache evidence requires a valid session token:

```
curl -k -H "Authorization: Bearer <session-token>" https://18-217-255-109.sslip.io/api/v1/roles
curl -k -H "Authorization: Bearer <session-token>" https://18-217-255-109.sslip.io/api/v1/roles
```

The first request should expose `X-Memory-Cache: MISS`; the second repeated request should expose `X-Memory-Cache: HIT`.

Automated Evidence

Automated validation is included in:

- `06Code/tests/integration/cache.test.js`
- `06Code/scripts/api-validation-suite.js`
- `06Code/python-analytics-api/tests/test_api.py`

Latest verified commands:

```
cd 06Code && npm test
Result: 5 test suites passed, 24 tests passed.

cd 06Code && npm run test:api:validation
Result: 114/114 API validation cases passed.

cd 06Code/python-analytics-api && .\.venv\Scripts\python.exe -m unittest
discover -s tests -v
Result: 9 tests passed.
```